

2 RELATED WORK

LLM Agents and MCP. The transformative capabilities demonstrated by LLMs (Jin et al., 2025) have enabled the development of LLM agents (Hua et al., 2024; Chae et al., 2025) capable of following natural language instructions, performing planning, and reasoning to solve complex tasks (Xu et al., 2024; Kojima et al., 2022; Lu et al., 2023; Shen et al., 2023). Through mechanisms such as function calling (Jarvis & Palermo, 2023), these agents can autonomously utilize external tools to interact with the real world (Schick et al., 2023; Qin et al., 2024; Chae et al., 2025; Gao et al., 2025; Abramovich et al., 2025). Furthermore, MCP standardizes agent-tool communication by providing a unified tool invocation interface (Antropic, 2024), which facilitates seamless interaction and significantly expands the operational scope of LLM agents (Zhang et al., 2025b; Qiu et al., 2025b;a; Fei et al., 2025). This unified protocol has rapidly emerged as foundational infrastructure for building advanced LLM agents (Hou et al., 2025).

MCP Specific Attack. The MCP ecosystem remains in its early stages, and the reliance of its decentralized architecture on remotely deployed servers introduces critical security vulnerabilities (Hou et al., 2025; Hasan et al., 2025; Song et al., 2025). Attackers can exploit tool information to manipulate LLM agents into invoking malicious tools or directly inject malicious instructions (Wang et al., 2025b; Labs, 2025). They can also supply parameters that deviate from a tool’s intended functionality, leading to unintended outcomes (Jing et al., 2025). Another effective attack vector involves fraudulent responses returned by tools invoked by the agent; these malicious outputs can induce the LLM agent to perform harmful actions (Guo et al., 2025). Furthermore, attacks described in Radosevich & Halloran (2025); Halloran (2025) embed malicious instructions within external data, which trigger an attack when retrieved by the LLM agent. MSB introduces a user-impersonating tool response attack and incorporates a comprehensive set of attacks spanning all stages of the MCP workflow.

Benchmarking Agents with Tool-invoking. Existing benchmarks for tool-invoking agents typically evaluate a narrow range of attacks, often only a single type (e.g., prompt injection) (Ruan et al., 2024; Gasmi et al., 2025; Zhan et al., 2024; Debenedetti et al., 2025), or consider evaluations involving multiple adversarial methods (Fu et al., 2025; Zhang et al., 2024). ASB (Zhang et al., 2025a) evaluates agent resilience against four attack categories, but its evaluation scope is limited to a simulated environment. Prior benchmarks are based on the function-calling paradigm and insufficient to cover the expanded attack surface presented by MCP. In contrast, MSB operates within a real-world dynamic environment where the agent is subjected to attacks spanning all stages of the MCP workflow, including several novel and challenging attacks introduced by MCP.

Existing MCP related benchmarks primarily focus on defining tool formats rather than probing the protocol’s security vulnerabilities (Fu et al., 2025). The recently proposed MCPTox (Wang et al., 2025a) is close in spirit to MSB but focuses solely on tool description injection attacks, utilizing LLM generated test cases. In contrast, MSB is designed to execute practical attacks targeting each stage of the MCP workflow in realistic scenarios, thereby providing a comprehensive evaluation of the diverse security vulnerabilities that MCP introduces against LLM agents.